

Fatma Deniz (born Imamoglu)

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Curriculum Vitae – May 2026

PROFESSIONAL EXPERIENCE

- | | |
|---|---------------------------|
| President
<i>Technische Universität Berlin – Berlin, Germany</i> | 04/2026 to Present |
| Vice President for Digitalisation and Sustainability
<i>Technische Universität Berlin – Berlin, Germany</i> | 04/2024 to 03/2026 |
| Full Professor (W3) Computer Science
<i>Faculty of Electrical Engineering and Computer Science, Technische Universität Berlin – Berlin, Germany</i>
Chair of Language and Communication in Biological and Artificial Systems | 04/2023 to Present |
| Senior Research Associate, Project Leader
<i>Faculty of Electrical Engineering and Computer Science, Technische Universität Berlin – Berlin, Germany</i>
Lead research project, "Language representations in bilinguals". | 2020 to 2023 |
| Assistant Project Scientist & Exceptional Principal Investigator
<i>Helen Wills Neuroscience Institute, University of California – Berkeley, CA, USA</i>
Established collaborative research effort, wrote CRCNS grant for five-year project. Received the grant as co-PI. | 2018 to 2023 |
| Data Science Fellow
<i>Berkeley Institute for Data Science (BIDS), University of California – Berkeley, CA, USA</i>
Led Reproducibility and Open Science working group for two years. Served as author and co-editor of "The Practice of Reproducible Research" book. Co-authored paper on using cloud-based technology to teach data science and paper on improving Internet authentication systems using findings from cognitive neuroscience. | 2014 to 2018 |
| Lecturer, Undergraduate Division, Data Science Education Program
<i>University of California – Berkeley, CA, USA</i>
Developed and taught course on Data Science for Cognitive Neuroscience based on Jupyter Notebooks and a cloud-platform in four sections over two semesters. Designed curriculum, created lesson plans and syllabi, and delivered lectures to undergraduate students. | 2016 to 2017 |
| Postdoctoral Scholar
<i>Helen Wills Neuroscience Institute, International Computer Science Institute, University of California – Berkeley, CA, USA</i>
Submitted and received two postdoctoral research grants. Studied semantic representations in human brains using naturalistic stimuli (reading and listening to narrative stories) and fMRI in collaboration with Professor Jack Gallant; published findings in Journal of Neuroscience and received extensive media coverage. | 2013 to 2018 |

EDUCATION

- Ph.D. (Dr. rer. nat.) in Computer Science (Magna Cum Laude), 2008 to 2013**
Technische Universität Berlin and Berlin Bernstein Center for Computational Neuroscience (BCCN) – Berlin, Germany
- *Dissertation: Visual Consciousness and Corticocortical Connectivity in the Human Brain (Advisor: Prof. John-Dylan Haynes & Prof. Christof Koch)*

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Diploma Thesis in Computational Neural Systems (Summa Cum Laude, 1.0*), 2007 to 2008

California Institute of Technology (Caltech) – Pasadena, CA

- Thesis: *Visual Saliency and Biological Inspired Text Detection* (Advisor: Prof. Christof Koch)

Diploma in Computer Science (Summa Cum Laude, 1.0*), 2001 to 2008

Technical University Munich (TUM) – Munich, Germany

- Minor in Computer Linguistics at the Ludwig Maximilian University of Munich (LMU) – Munich, Germany

High-School, 1997 to 2001

Bursa Kiz Lisesi – Bursa, Turkey

- High-school graduation (Ranked 1st)

* Grading Scheme in Germany: 1.0–1.5 very good, 1.6–2.5 good, 2.6–3.5 satisfactory, 3.6–4.0 sufficient, 5.0 fail

GRANTS & FELLOWSHIPS (LAST FIVE YEARS)

German Research Foundation (DFG), Collaborative Research Center, SFB 1404/2, Project Grant, co-PI. EUR 638 400. **2024-2027.**

European Research Council (ERC), Starting Grant. Research Grant, PI. EUR 1 499 885. **2023-2027.**

UCSF, UC Berkeley Schwab Dyslexia and Cognitive Diversity Center. Research Grant, Co-PI. USD 105 500. **2022-2023.**

Bundesministerium für Bildung und Forschung, Federal Grant for a 3-Tesla Magnetic Resonance Imaging Scanner, Total: EUR 3 944 144. **2021.**

USA-Germany Collaborative Research in Computational Neuroscience. Research Grant, Co-PI. Total: EUR 1 773 817; NSF (Award Number: 1912373) USD 920 079 plus BMBF (Förderkennzeichen: 01GQ1906): EUR 927 345. **2020-2025.**

PUBLICATIONS

Books

1. Kitzes J., Turek D., **Deniz F.**, editors, *The Practice of Reproducible Research: Case Studies in Data Science*, UC Press. 2017. Online book available at <https://www.practicereproduciblesearch.org/>.

Book Chapters

1. **Deniz F.**, "pyMooney: Generating a Database of Two-Tone, Mooney Images." In *The Practice of Reproducible Research: Case Studies in Data Science*. Kitzes, J., Turek D., **Deniz F.**, editors. UC Press. 2017
2. Turek D., **Deniz F.**, "Introducing the Case Studies." In *The Practice of Reproducible Research: Case Studies in Data Science*. Kitzes, J., Turek D., **Deniz F.**, editors. UC Press. 2017

Journal Publications and Conference Proceedings

1. Chen C., Gong X. L., Tseng C., Klein D., Gallant J.L., **Deniz F.**, Bilingual language processing relies on shared semantic representations that are modulated by each language, *PNAS*, 123 (9) e2503721123, 2026.
2. Negi A.*, Oota S. R.*, Nunez-Elizalde A., Gupta M., **Deniz F.**, Brain-Informed Fine-Tuning for Improved Multilingual Understanding in Language Models, *NeurIPS*, 2025.
3. Siegel NT., Kainmueller D., **Deniz F.**, Ritter K., Schulz MA. Do Transformers and CNNs Learn Different Concepts of Brain Age? *Hum Brain Mapp.* Jun 1;46(8):e70243. doi: 10.1002/hbm.70243, 2025.
4. Chen C., Dupre de la Tour T., Gallant J.L., Klein D., **Deniz F.**, The cortical representation of language timescales is shared between reading and listening. *Nature Communications Biology*, 7, 284, 2024.
5. Oota S. R., Celik E., **Deniz F.**, Toneva M., Speech language models lack important brain-relevant semantics, *Annual Meeting of the Association for Computational Linguistics*, 8503–8528, 2024.
6. Gong X. L., Huth A. G., **Deniz F.**, Johnson K., Gallant J. L., Theunissen, F. E., Phonemic segmentation of narrative speech in human cerebral cortex. *Nature Communications*, 14(1), 4309, 2023.
7. **Deniz F.***, Tseng C.*, Wehbe L., Dupre la Tour T., Gallant J. L., Semantic representations during language comprehension are affected by context. *Journal of Neuroscience*, 43(17), 3144–3158, Featured Research March 28th, 2023.

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8. Lamarre M., Chen C., **Deniz F.**, Attention weights accurately predict language representations in the brain, In Association for Computational Linguistics: EMNLP, Empirical Methods in Natural Language Processing, 2022.
9. Popham S., Huth A. G., Bilenko N. Y., **Deniz F.**, Gao, S. J., Nunez-Elizalde A., Gallant J. L., Visual and linguistic semantic representations are aligned at the border of human visual cortex. *Nature Neuroscience* 24, 1628-1636. 2021.
10. Ramakrishnan K. and **Deniz F.**, Complementarity of information in word-embedding and brain representations in distinguishing between concrete and abstract words. In Proceedings of NAACL HLT Workshop on Cognitive Modeling and Computational Linguistics CMCL. 2021.
11. **Deniz F.**, Nunez-Elizalde A.O., Huth A. G., Gallant J. L., The representation of semantic information across human cerebral cortex during listening versus reading is invariant to stimulus modality. *Journal of Neuroscience*, 39 (39) 7722-7736. 2019.
12. Castelluccia C., Duermuth M., Golla M., **Deniz F.**, MooneyAuth: Towards Practical Implicit Memory-based Authentication. In Network and Distributed System Security Symposium (NDSS). 2017.
13. Holdgraf C., Culich A., Rokem A., **Deniz F.**, Ushuzima D., Alegro M., Portable learning environments for hands-on computational instruction: Using container- and cloud-based technology to teach data science. In Proceedings of the Practice and Experience in Advanced Research Computing on Sustainability, Success and Impact (p. 32). ACM. 2017.
14. Kizilirmak J. M., Gomes da Silva J. G., **Imamoglu F.**, Richardson-Klavehn A., Generation and the subjective feeling of "aha!": Independent contributions to learning from insight. *Psychological research* 80 (6), 1059-1074. 2015.
15. Castelluccia C., Duermuth M., **Imamoglu F.**, Learning from Neuroscience to Improve Internet Security. *ERCIM News*, 99:46. 2014.
16. **Imamoglu F.**, Heinzle J., Imfeld A., Haynes J.-D., Activity in high-level brain regions reflects visibility of low-level stimuli. *Neuroimage*, 102:688-694. 2014.
17. **Imamoglu F.**, Kahnt T., Koch C., Haynes J.-D., Changes in functional connectivity support conscious object recognition. *Neuroimage*, 63:1909-17. 2012.

Manuscripts on arXiv, in Preparation and Submitted

1. Negi A., Tseng C., Nunez-Elizalde A. O., Gong L. X., **Deniz F.**, Optimizing Language Model Embeddings to Voxel Activity Improves Brain Activity Predictions, *bioRxiv* 2025.09.18.676935; doi: <https://doi.org/10.1101/2025.09.18.676935>
2. Visconti di Oleggio Castello M., **Deniz F.**, Tom Dupré la Tour, Gallant J. L., Encoding models in functional magnetic resonance imaging: the Voxelwise Encoding Model framework. *PsyArXiv*, preprint, 2025.
3. Wu M.*, **Deniz F.***, Prenger R., Gallant J. L., The unified maximum a posteriori (MAP) framework for neuronal system identification (* First authorship shared), *arXiv*, <https://arxiv.org/abs/1811.01043>.

PATENT APPLICATION

Imamoglu F., Golla M., Duermuth M., Castelluccia C., Method for authenticating a user, associate system and computer program, international publication number: WO2015193389. 2015.

SELECTED MEDIA COVERAGE

Zeit Online. „[Ist Hörbuchhören genauso viel wert wie Lesen?](#)“.

Tagesspiegel. „[Hören oder Lesen?: Ein Blick ins Gehirn zeigt, wann mehr vom Text hängen bleibt](#)“.

MIT Technology Review. „[A map of the brain could help to guess what you're reading](#).“

Heise Magazine Technology Review. „[Hirnsan verrät, was Menschen lesen](#).“

Discover Magazine. „[Audiobooks or Reading? To Our Brains, It Doesn't Matter](#).“

The Daily Californian. „[UC Berkeley study finds human brains similarly stimulated by reading, listening](#).“

The Telegraph: „[End of audiobook snobbery as scientists find reading and listening activates the same parts of the brain](#).“